

Question			Marking details	Marks available				
				AO1	AO2	AO3	Total	Prac
	(b)	(i)	Correct use of $P = A\sigma T^4$ for either Sirius A or Canopus or both i.e. Canopus: $P = 4 \times \pi \times (4.97 \times 10^{10})^2 \times 5.67 \times 10^{-8} \times (7250)^4$ Sirius A: $P = 4 \times \pi \times (1.19 \times 10^9)^2 \times 5.67 \times 10^{-8} \times (10000)^4$ (1) Canopus: $P = 4.9 \times 10^{30}$ [W] (1) Sirius A: $P = 1.0 \times 10^{28}$ [W] (1) Accept powers of 10 error in answers omission of 4 deduct 1 mark $\frac{4.9 \times 10^{30}}{1.0 \times 10^{28}}$ (≈ 500) shown (1) Alternative: Attempt at $P = A\sigma T^4$ used in ratio (1) e.g. $\frac{L_C}{L_S} = \frac{49.7 \times 7250^4}{1.192 \times 10000^4} \approx 500$ (1)	1	1 1		4	4
		(ii)	$I = \frac{1.0 \times 10^{28}}{4\pi (8.15 \times 10^{16})^2}$ substitution (1) ecf $I = 1.19 \times 10^{-7}$ [W m⁻²] (1) Accept 1.21×10^{-7} [W m⁻²]	1	1		2	2
		(iii)	Canopus is further away from earth or Sirius A is closer to earth (1) Intensity reaching earth $\propto \frac{1}{R^2}$ or P from star spread out over greater surface area (1) Accept intensity equation Don't accept intensity $\propto \frac{1}{\text{distance}}$ or because of the inverse square law		2		2	
			Question 3 total	2	7	4	13	0